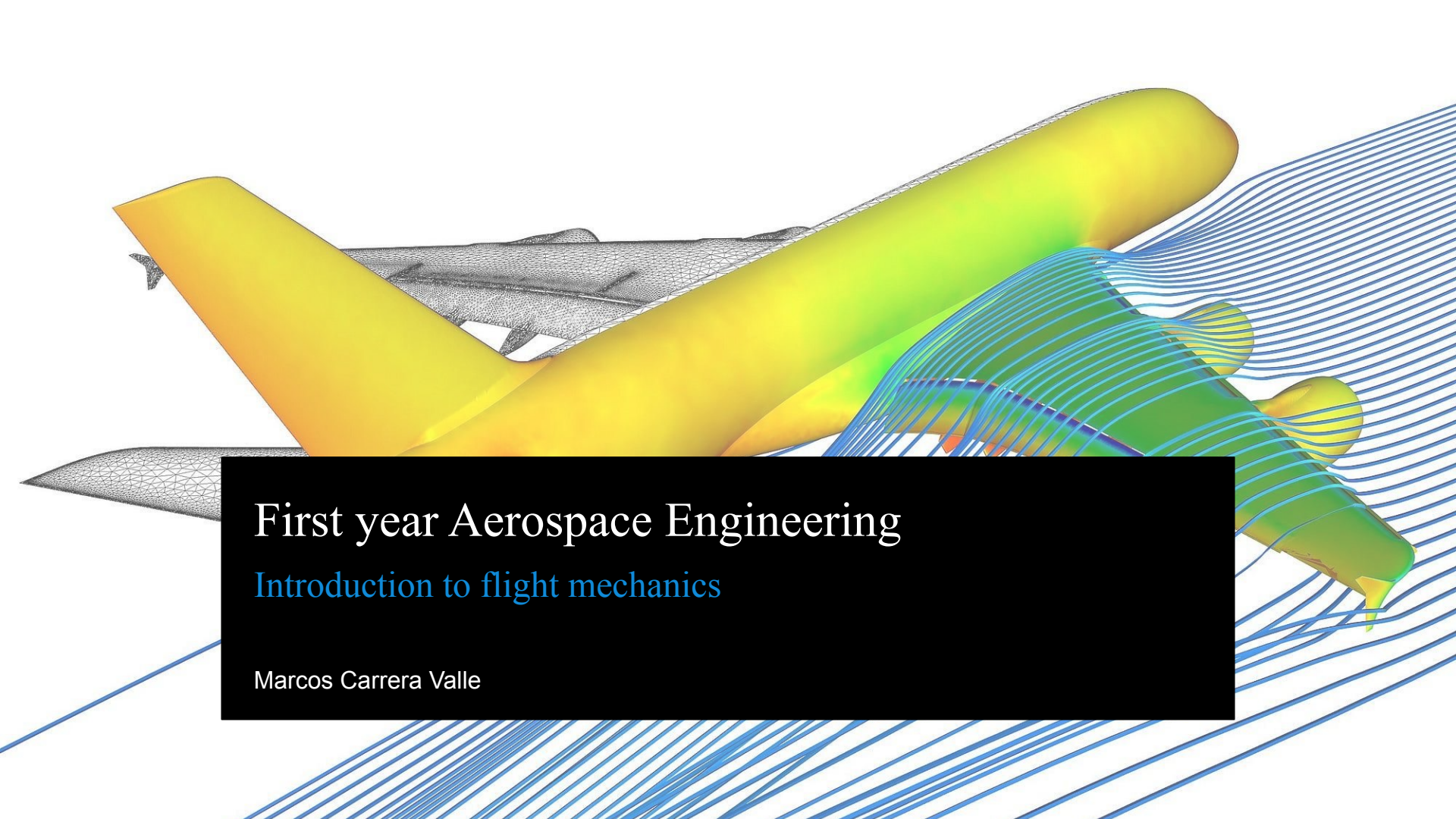




First year Aerospace Engineering

Introduction to flight mechanics

Marcos Carrera Valle

What is Flight Mechanics?

Basically it is the study that measures the performance of the aircraft

How fast, slow, high, low, long an aircraft can go



Equations of motion

Equation of motion are derived from Newton's second law ($F = m \cdot a$)
So we need to find the forces acting on the aircraft and the accelerations of the aircraft

First we need to define the frames of reference

Free body diagram

g: Earth axis system

E: Moving earth axis system

B: Body axis system

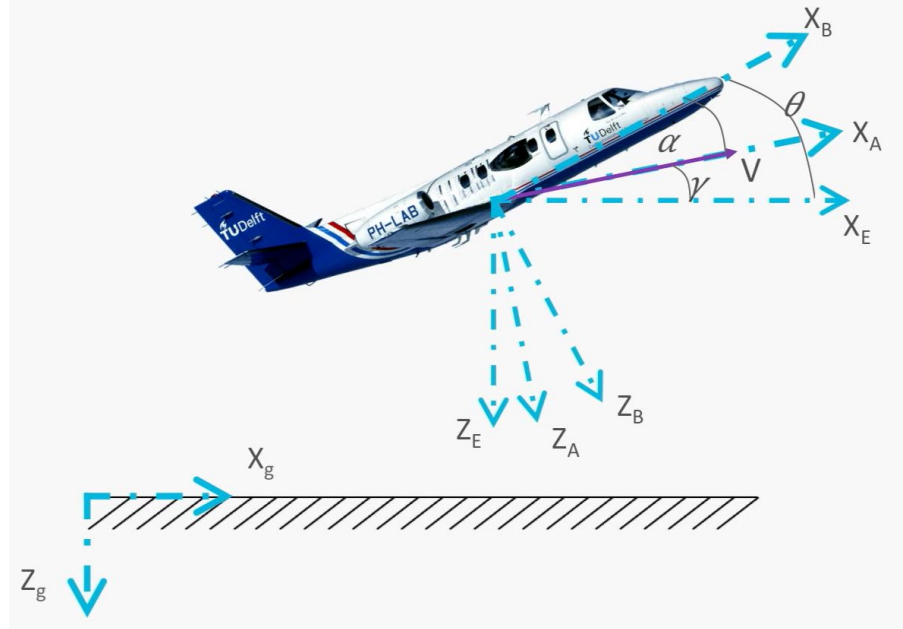
A: Air path axis system

α : Angle of attack (body w.r.t to airflow)

γ : Flight path angle (airflow w.r.t to horizon)

θ : Pitch angle (body w.r.t to horizon)

This works assuming the earth is flat and isn't rotating



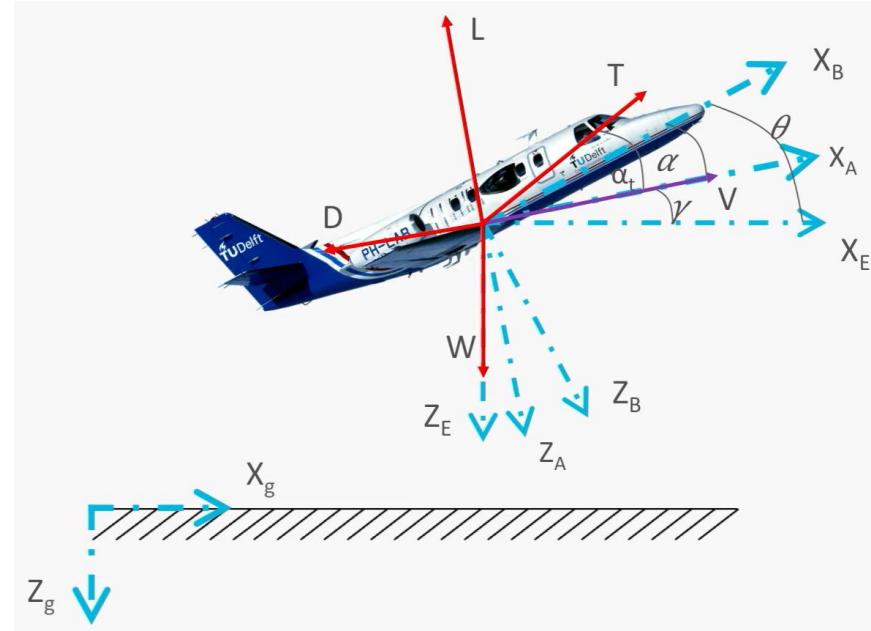
Forces

Lift acts perpendicular to airflow path (X_A)

Drag acts parallel to airflow path (X_A)

Weight acts towards the center of the earth, so parallel to Z_E

Thrust acts at an angle of α_T (thrust angle of attack), w.r.t to the airflow path X_A
(Usually this is the same as α)



Accelerations

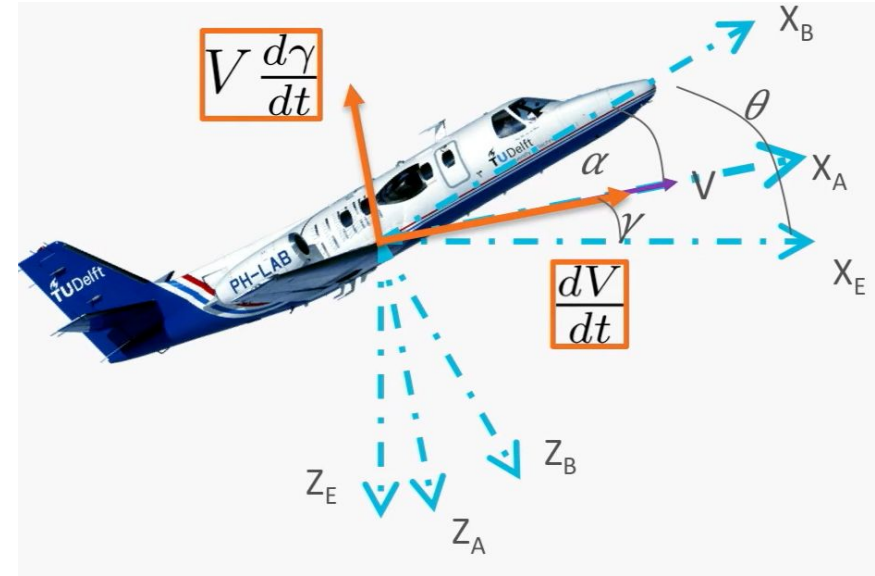
From experience it has been shown that the most convenient accelerations for flight mechanics are taken w.r.t the air path axis system (X_A, Z_A)

Parallel to V , we have dV/dt
Perpendicular to V , we have:

$$a = \frac{V^2}{R} \quad (1) \quad \omega R = V \quad (2) \quad \omega = \frac{d\gamma}{dt} \quad (3)$$

simplifies to

$$a = V \frac{d\gamma}{dt}$$



Equations of motion

Using $\text{sum of forces} = m \cdot a$ and some assumptions, we get:

Equations of motion

Using sum of forces = $m \cdot a$ and some assumptions, we get:

$$\sum F_{\parallel V} : \frac{W}{g} \frac{dV}{dt} = T \cos \alpha_t - D - W \sin \gamma$$
$$\sum F_{\perp V} : \frac{W}{g} V \frac{d\gamma}{dt} = L - W \cos \gamma + T \sin \alpha_t$$

The equations of motion!

Variables

To describe the motion of the plane we have to use 2 **state** variables:

Flight path angle, γ

Velocity, V

With these state variables we can describe the speed and direction of the plane

We will calculate these variables at a given time using

1 independent variable, t (time)

Other variables depending on the pilot and aircraft (T , L , W , D , α_T)

Drag

Drag is an important measurement for flight mechanics, as we want to minimize it

Drag depends on parasitic (zero-lift drag), C_{D_0}
and lift drag,

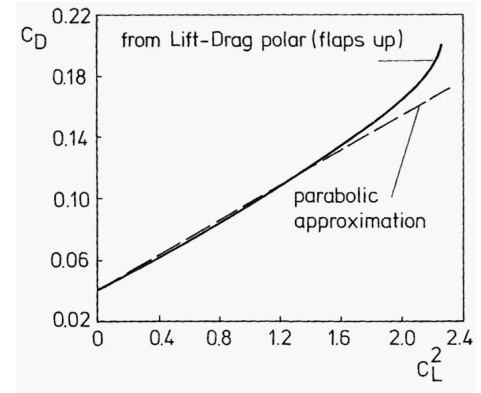
Such that: $\frac{C_L^2}{\pi A \phi}$

$$C_D = C_{D_0} + \frac{C_L^2}{\pi A \phi}$$

Drag polar

$$C_D = C_{D_0} + \frac{C_L^2}{\pi A \phi}$$

This is called the drag polar



In essence a parabolic equation: $y = ax^2 + b$

To be more accurate: $y = ax^2 + bx + c$

Where in this case $x = C_L$ and $c = C_{D_0}$

Drag

Now we can define drag as $D = \frac{1}{2} * \rho * V^2 * c_D * S$

Let's derive this in terms of c_L and of V

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Drag

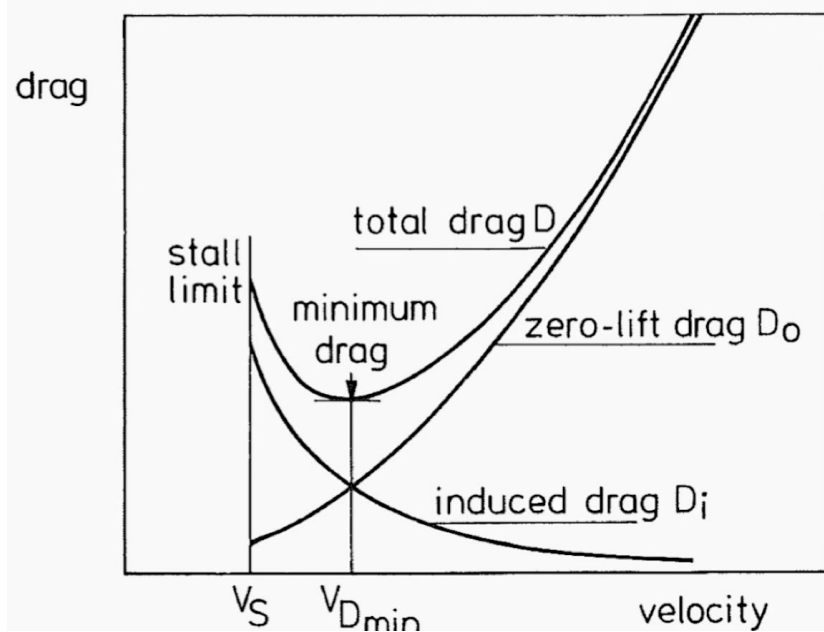
Now we can define drag as $D = \frac{1}{2} * \rho * V^2 * c_D$

Let's derive this in terms of c_L and of V

Eventually we see that $D = f(V^2) + f(1/V^2)$

So a part of drag increases with V and another part decreases with V ?

Drag



So at low airspeeds, we actually decrease total drag by increasing speed!

Fast aircraft have to deal with zero-lift drag, so they are slim to reduce parasitic drag, and have low aspect ratio

Slow aircraft have to deal with (lift) induced drag, so they have a high aspect ratio

$$\frac{C_L^2}{\pi A \phi}$$

Aircraft morphology



$$\frac{C_L^2}{\pi A \phi}$$



Propulsion

Obviously we want to move forward, so we have the propulsion system

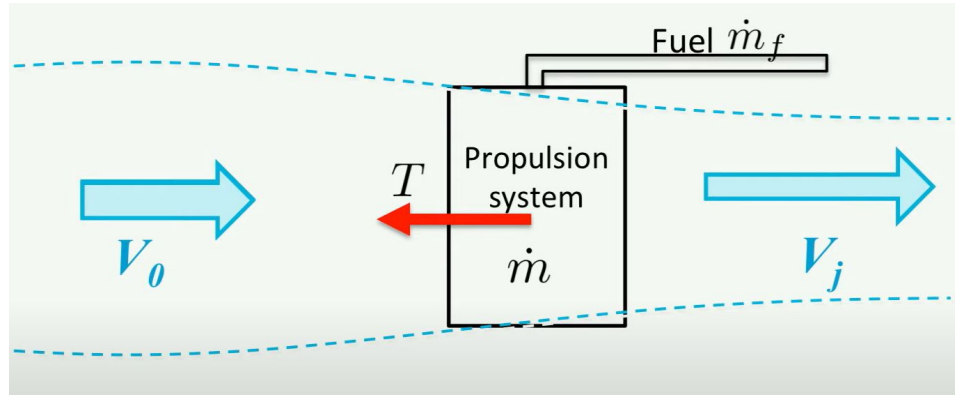
How do engines work?

They push air mass backwards, pushing the aircraft forward (conservation of momentum)

Momentum before = Momentum after

Thrust

In the engine nozzle, air enters at a velocity V_0 and a given mass flow (kg/s)
In the engine, fuel is burnt, given the fuel-air mixture kinetic energy
The fuel air-mixture then leaves the engine at a velocity V_j which is higher than V_0 and with a new mass flow



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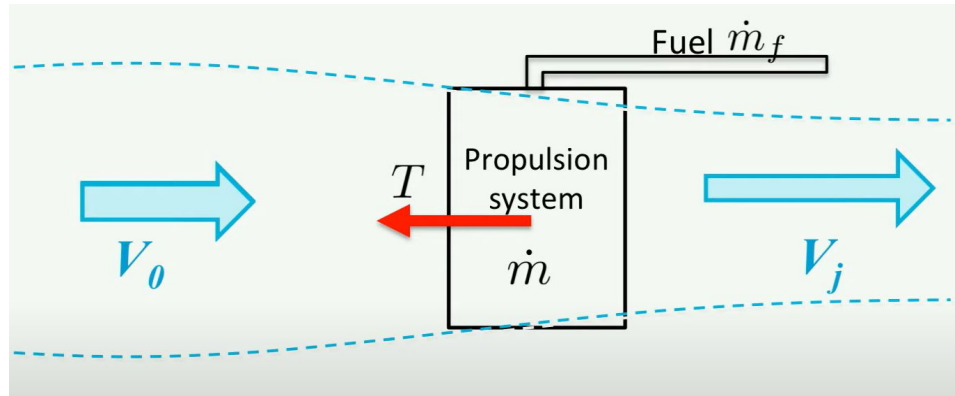
From the momentum equation, Thrust = Change in momentum rate

Momentum = mass * velocity, Momentum rate = mass flow * velocity

Thrust

Momentum rate before = $V_0 * m$

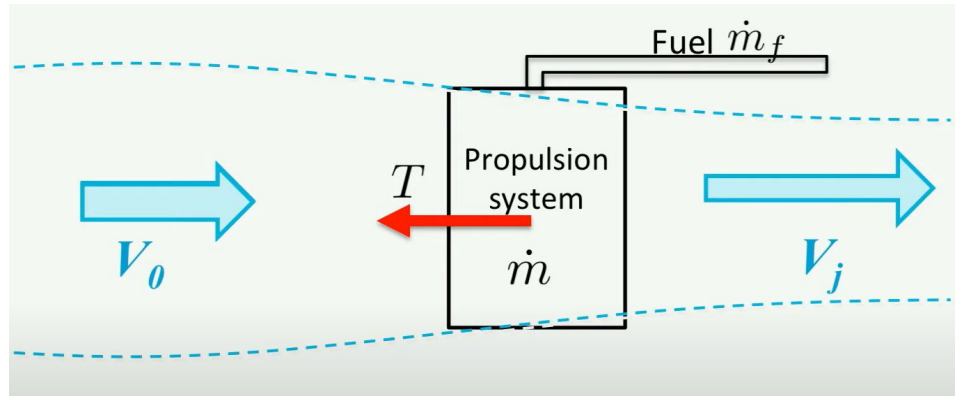
Momentum rate after = $V_j * (m + m_f)$



Thrust

After some simplification we get:

$$T = \dot{m} \cdot (V_j - V_0)$$



Thrust

An engine has a mass flow of 15 kg/s, and produces a thrust of 2500 N

If the velocity of the aircraft is 240 m/s, what is the jet velocity?

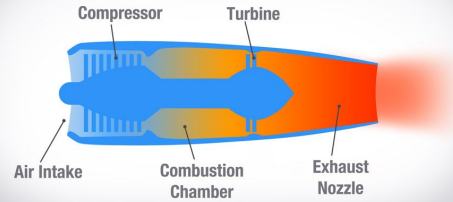
Engine types

We can have various types of engines, most notably jet and propeller engines



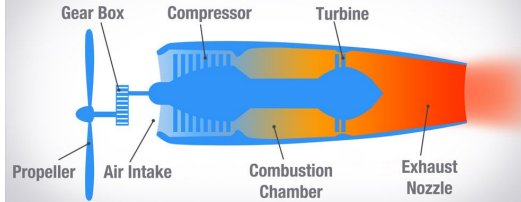
Engine types

Turbojet Engine



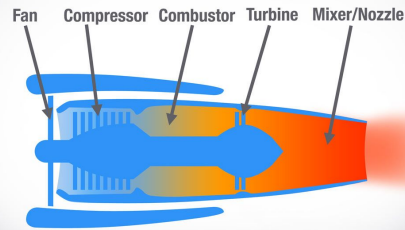
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Turboprop Engine



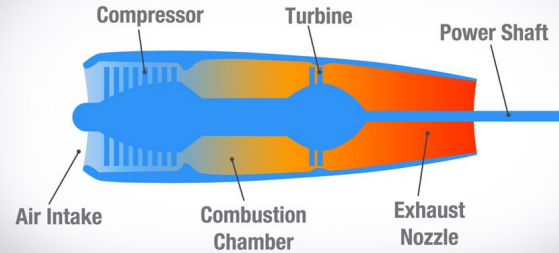
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Parts Of A Turbofan



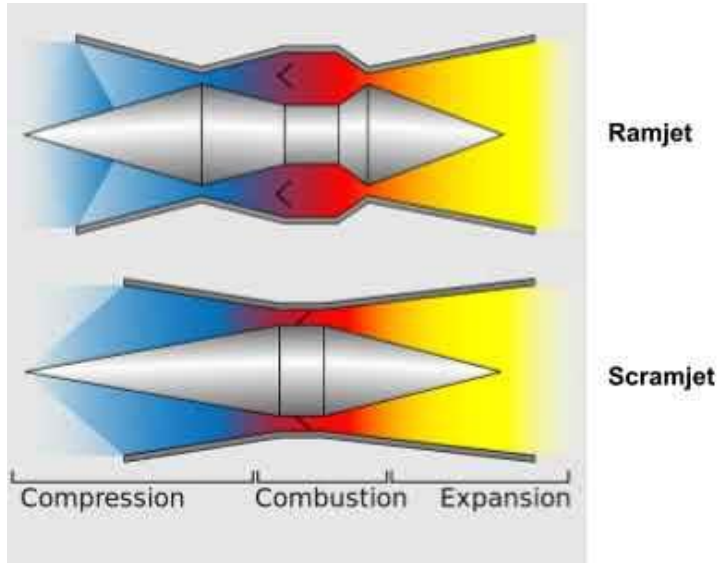
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Turboshaft Engine



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Engine types



Engine types

We can have various types of engines, most notably jet and propeller engines

Jet engines give small masses of air a lot of velocity increase

Propeller engines give large masses of air a small velocity increase



Engine types

To see when should we use each engine we have to look at efficiency



Efficiency

Efficiency is the useful power / total power used

Power = Energy over time, and energy is force * distance

So useful power/ power available is given by: $P_a = TV$

Efficiency

Efficiency is the useful power / total power used

Power = Energy over time, and energy is force * distance

So useful power/ power available is given by: $P_a = TV$

To achieve this power, we burn fuel. The power released by burning the fuel is given by

$$Q = \dot{m}_f H$$

$$H = \text{Amount of energy per unit of fuel}$$

Efficiency - jet engines

The total efficiency of an engine is then given by

$$\eta_t = \frac{P_a}{Q}$$

We introduce a new term for jet engines, the jet power. It is the kinetic energy rate behind the aircraft - the kinetic energy rate in front of the aircraft

Jet power:

$$P_j = \frac{1}{2} \dot{m} V_j^2 - \frac{1}{2} \dot{m} V_0^2$$

Efficiency - jet engines

We can use this term to divide the total engine efficiency into 2 parts, thermal efficiency and propulsive efficiency, where

$$\eta_t = \eta_{th} \cdot \eta_j \quad \text{with:} \quad \eta_{th} = \frac{P_j}{Q}$$
$$\eta_j = \frac{P_a}{P_j}$$

Propulsive efficiency

An aircraft is equipped with an engine that provides 1000 N of thrust, with a mass flow of 10 kg/s

If the velocity of the aircraft is 100 m/s, what is the propulsive efficiency η_j ?

Propulsive efficiency

An aircraft is equipped with an engine that provides 1000 N of thrust, with a mass flow of 10 kg/s

If the velocity of the aircraft is 200 m/s, what is the propulsive efficiency η_j ?

Propulsive efficiency

As you can see, the propulsive efficiency can drastically change with flight velocity!

That explains why jet engines, which have high jet velocities, fly at high velocities: because they are much more efficient at high velocities than at low velocities

Thrust as a function of V

Recall that $T = \dot{m} \cdot (V_j - V_0)$

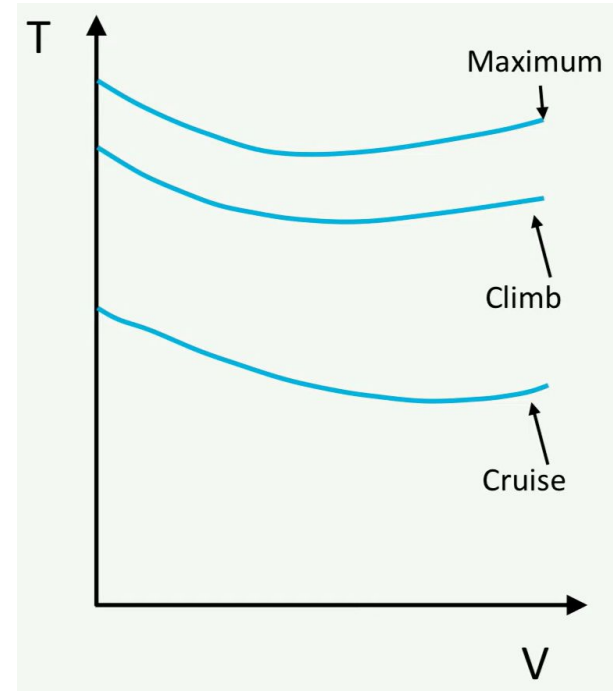
What happens if we increase V_0 ?

Thrust as a function of V

Recall that $T = \dot{m} \cdot (V_j - V_0)$

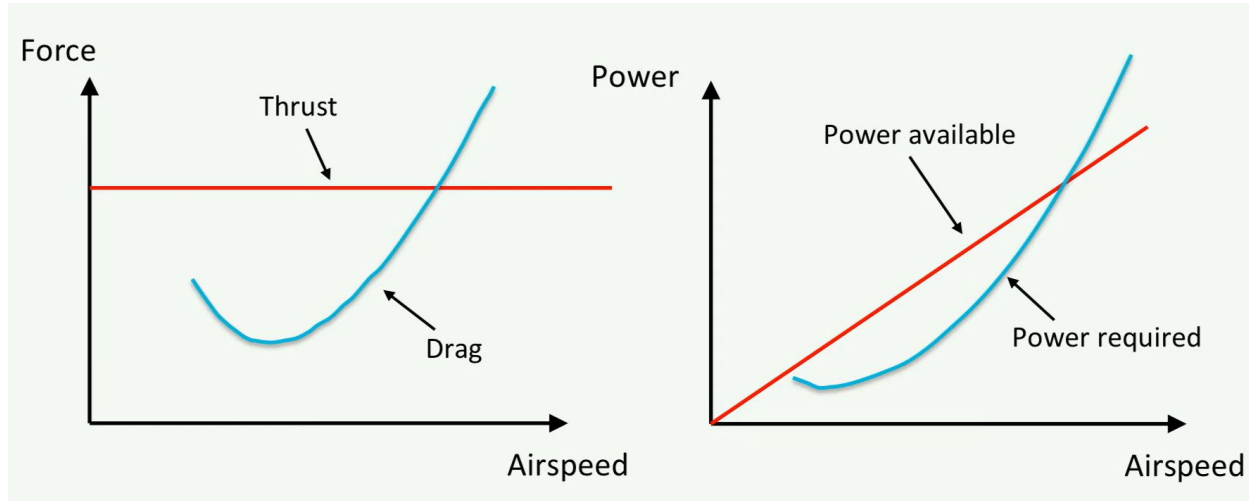
What happens if we increase V_0 ?

Thrust remains almost constant with changing V
(for a given flight altitude and throttle setting)



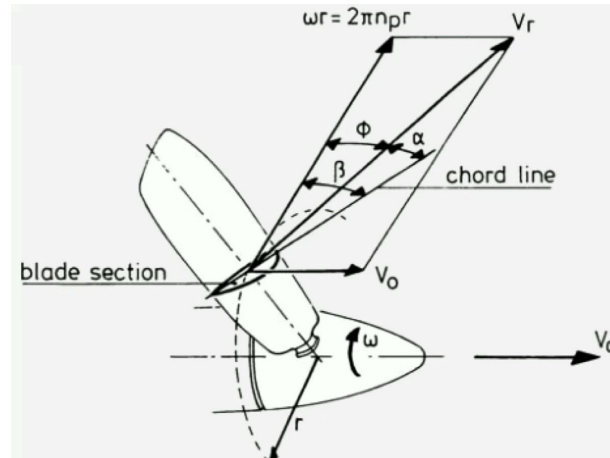
F-V and P-V diagrams - jet engines

We can plot Force - Velocity and Power - Velocity diagrams, fundamental for flight mechanics analysis



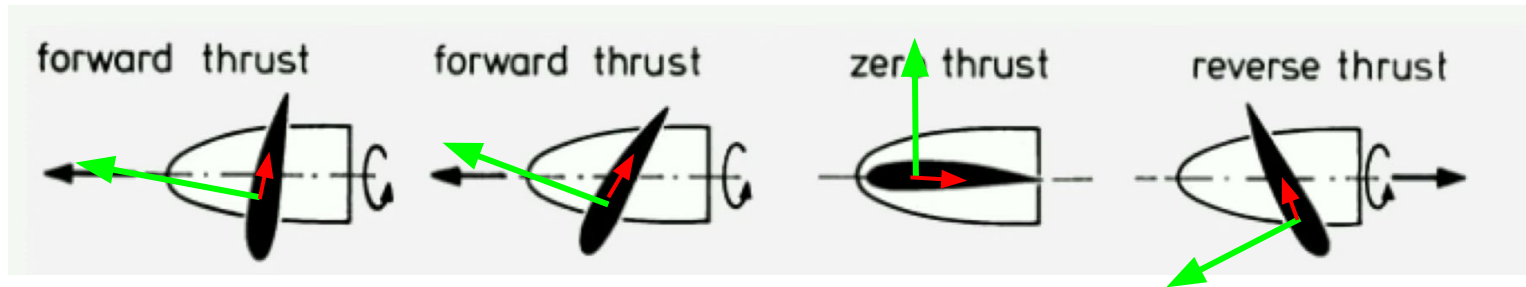
Propeller engines

Now onto propeller engines: how do they work?



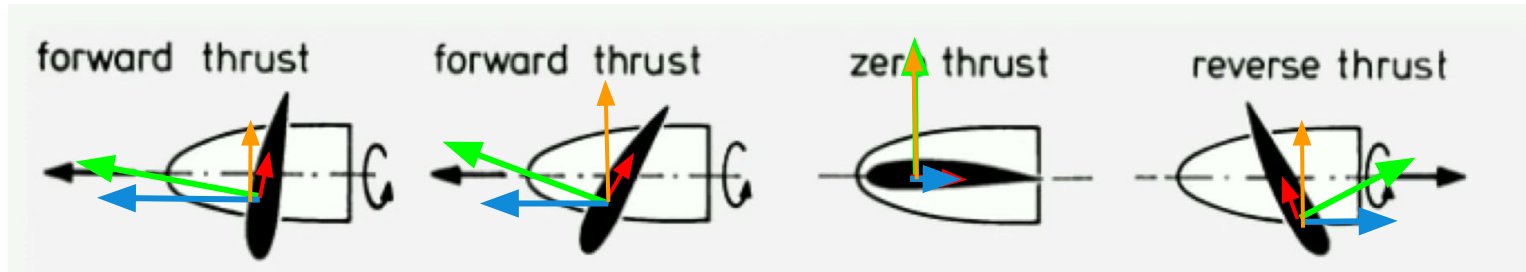
Propeller engines

The blades act like 'mini wings', that generate lift and drag



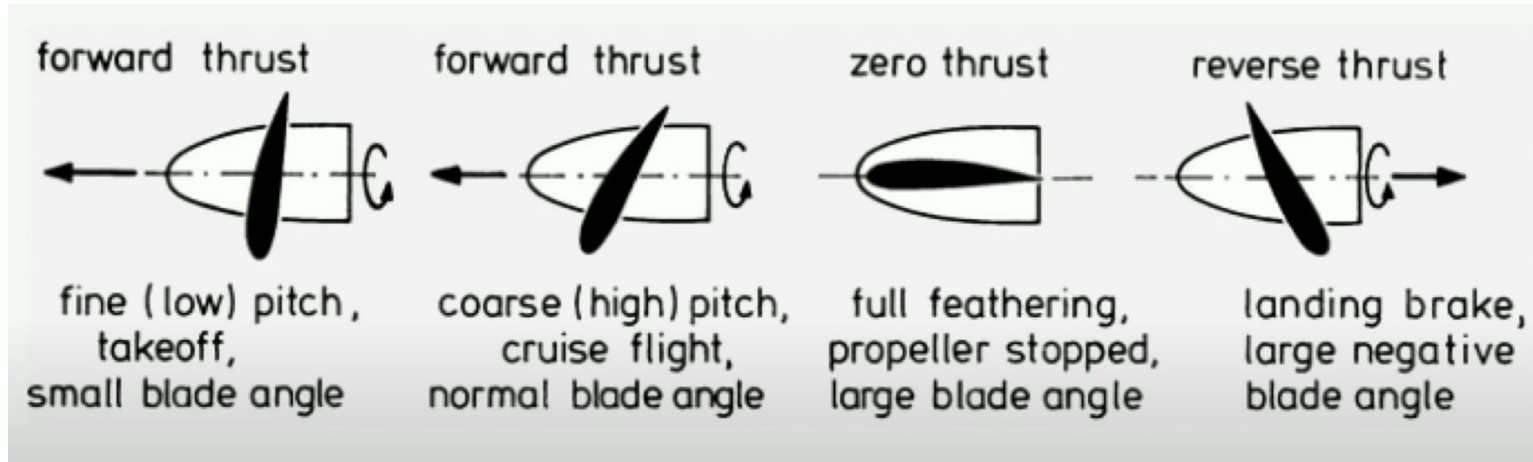
Propeller engines

The lift and drag component perpendicular to the airspeed generate **Thrust**
The lift and drag component perpendicular to the airspeed generate a **torque** in the shaft that the engine must overcome



Propeller engines

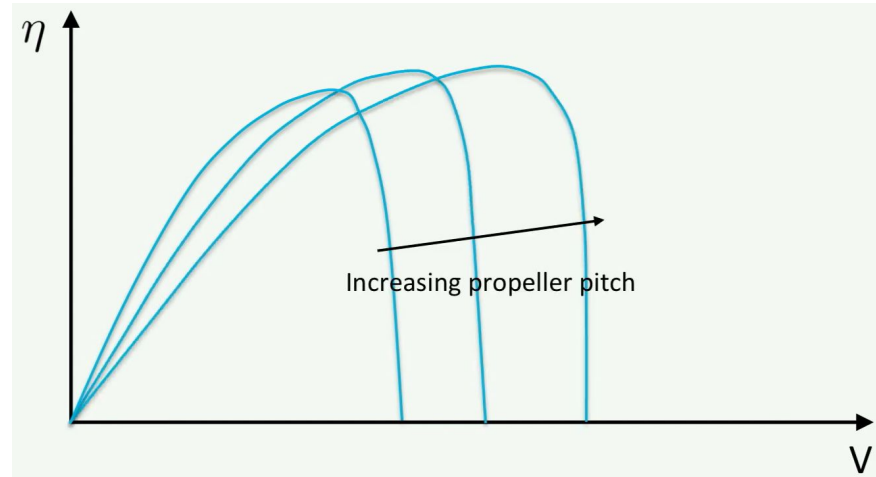
We can change the pitch geometry depending on the flight speed



Propeller engines

At different velocities, we can “maintain” the propulsive efficiency by varying the propeller pitch

So we can assume that propulsive efficiency is constant with velocity



Propeller engines

For a propeller, the propulsive efficiency is the available power P_a , divide by the power delivered to the shaft, (shaft power, P_{br})

$$\eta_j = \frac{P_a}{P_{br}}$$

Now if shaft power is kept constant, and efficiency is kept constant, we can say that available power is constant

Propeller engines

For a propeller, the propulsive efficiency is the available power P_a , divide by the power delivered to the shaft, (shaft power, P_{br})

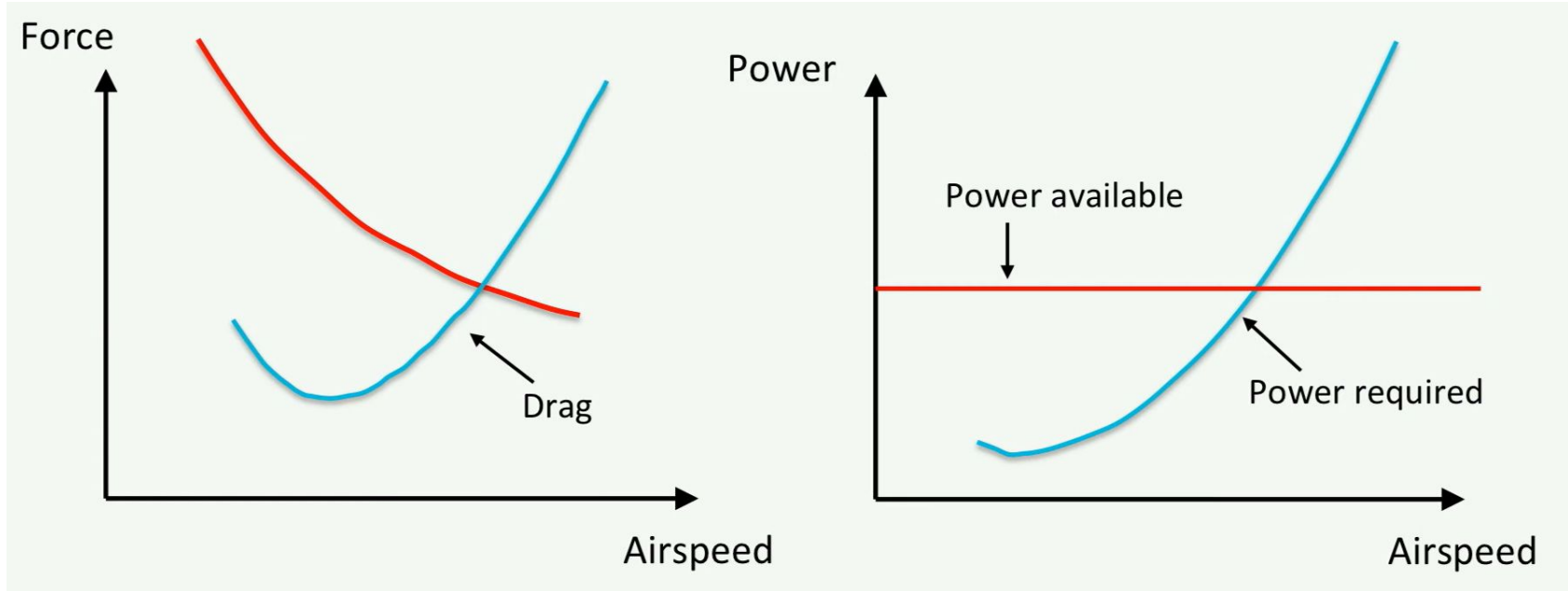
$$\eta_j = \frac{P_a}{P_{br}}$$

Now if shaft power is kept constant, and efficiency is kept constant, we can say that available power is constant

As thrust is available power / velocity, we can also say that thrust is inversely proportional to velocity

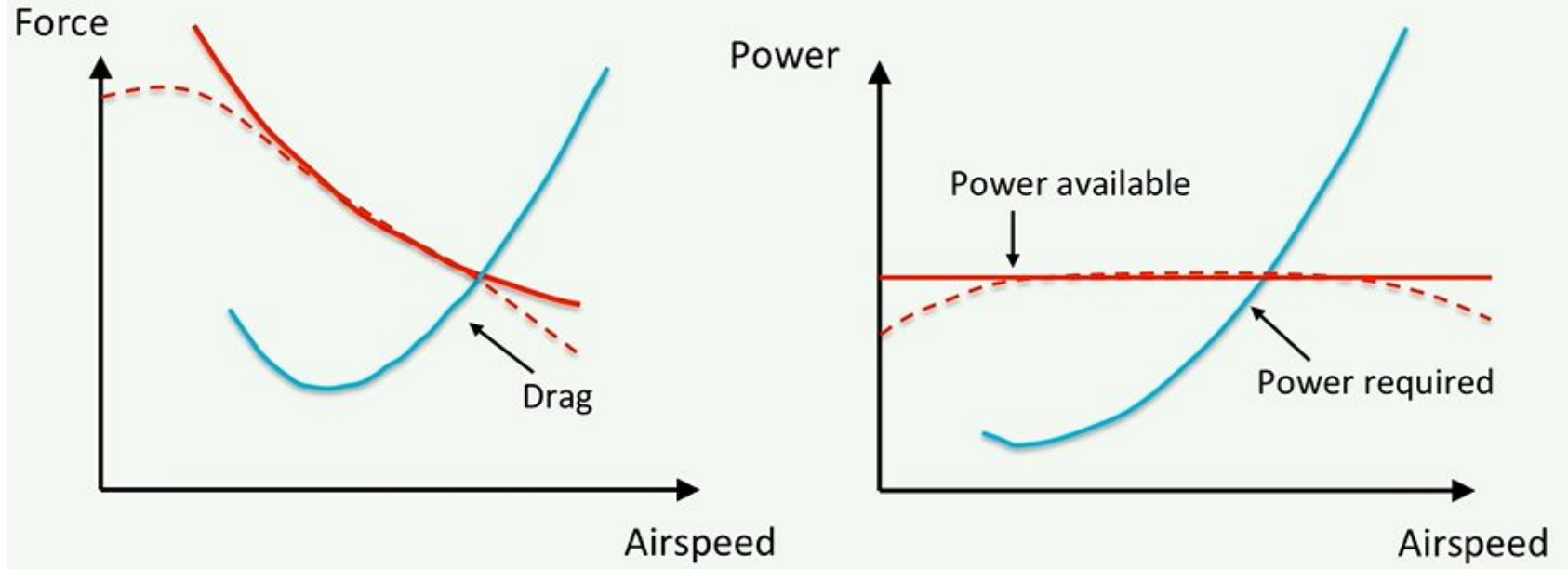
F-V and P-V diagrams - propeller engines

This is just theoretical behaviour, in reality...



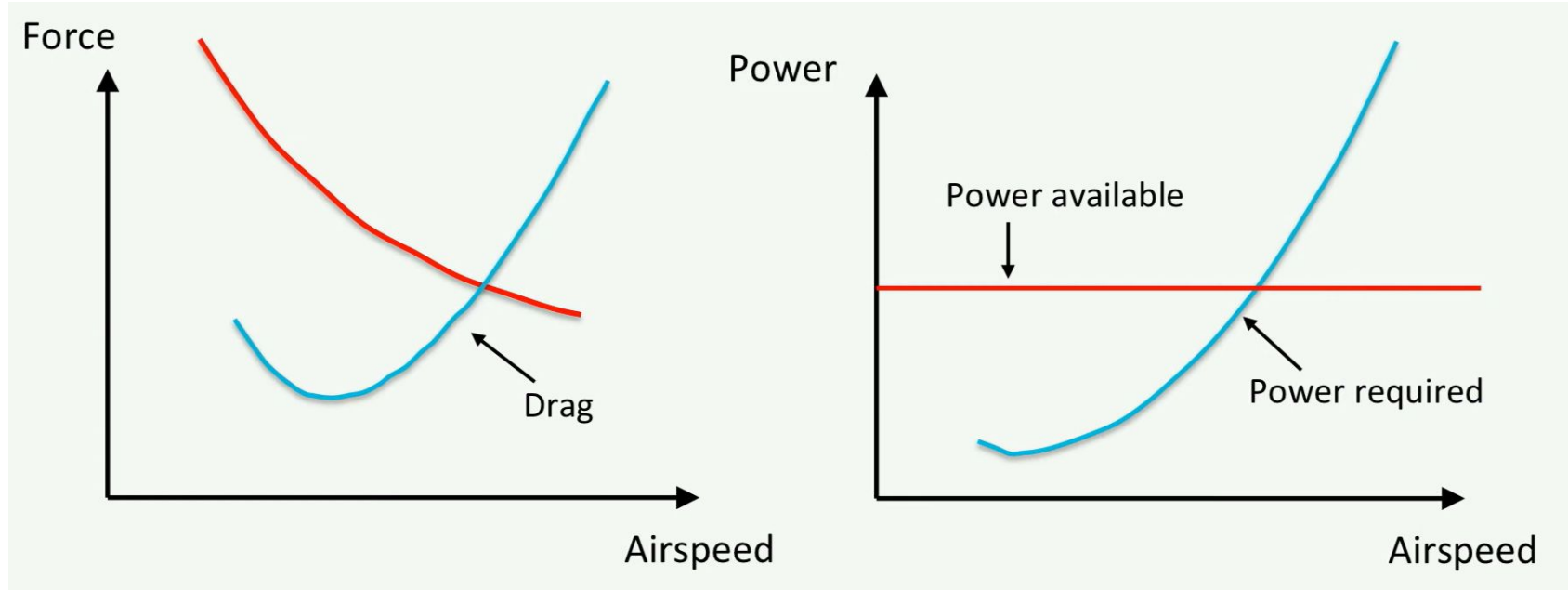
F-V and P-V diagrams - propeller engines

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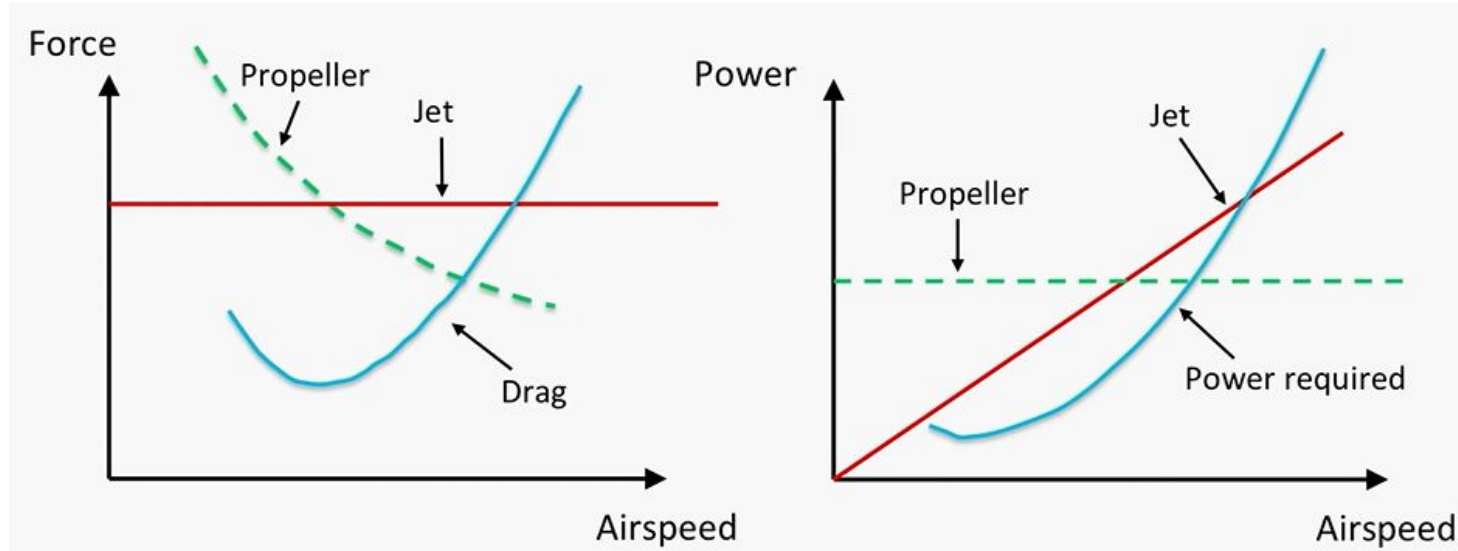


F-V and P-V diagrams - propeller engines

But we can assume this for calculations



Performance Diagrams - Simplified Engines



Exercise 1

Propulsion problem 1 (from nov 2013 exam)

Please find below some relevant data for the Boeing 747 airliner:

Number of engines:	4
Wing surface:	$S = 511 \text{ m}^2$
Diameter of engine inlet:	$d = 2.38 \text{ m}$
Oswald efficiency:	$e = 0.70$
Span:	$b = 60 \text{ m}$
Mass/Weight:	$m = 350000 \text{ kg}$
Cruise altitude:	$h = 35000 \text{ ft}$ (1 ft = 0.3048 m)
Cruise speed:	$V_{TAS} = 480 \text{ kts}$ (1 kts = 0.514444 m/s)
Zero-lift drag coefficient:	$C_{D0} = 0.022$



$$C_D = C_{D0} + \frac{C_L^2}{\pi A e}$$

Exercise 1

For the International Standard Atmosphere (ISA) please use:

Gravity acceleration:	g	$= 9.80665 \text{ m/s}^2$
Gas constant for air:	R	$= 287.00 \text{ J/ kgK}$
Sea-level pressure:	p_0	$= 101325 \text{ Pa}$
Sea-level temperature:	T_0	$= 15 \text{ deg Celsius}$

Sea-level density: $\rho_0 = 1.225 \text{ kg/m}^3$

For $h \leq 11 \text{ km}$ Temperature gradient: $a = - 6.50 \text{ K/km}$

For $11 \text{ km} \leq h \leq 20 \text{ km}$: isothermal layer

$$\frac{p_1}{p_0} = \left(\frac{T_1}{T_0} \right)^{-\frac{g_0}{aR}}$$

Exercise 1

$$\text{density} = \rho = 0.3796$$

- b) Calculate the required available power **per engine** for the cruise at the given cruise speed and altitude.
- c) Calculate the mass flow in **one engine** for the given cruise altitude and cruise speed
- d) Calculate the jet exhaust speed for the given cruise altitude and cruise speed
The propulsive efficiency of a jet engine is defined as:

$$\eta_{jet} = \frac{\text{Power available for thrust}}{\text{Change in kinetic energy of jet for the mass flow}}$$

- e) Calculate the propulsive efficiency of the jet engine of the 747 for this cruise flight

Exercise 2

Let's talk about the beautiful Boeing 747.

Here are a few characteristics:

$W = 3,300,000 \text{ N}$

Oswald efficiency factor, $e = 0.9$

$C_{D,0} = 0.018$

Wing area = 511 m^2

Span = 60 m

It flies at 926 km/h , at flight level FL 35000, where $\rho = 0.379597$



Exercise 2

What is the total drag?

Which drag component is dominant, zero lift drag or lift induced drag?

If the aircraft increases its velocity, will the drag increase or decrease? Why?

Exercise 2

This aircraft has **4** engines.

Each engine has an inlet diameter of 2.6 m.

What is the inlet area? What is the mass flow of this engine?

Exercise 2

This aircraft has **4** engines.

Each engine has an inlet diameter of 2.6 m.

What is the inlet area? What is the mass flow of **1** engine?

If you assume $T = D$, what is the jet velocity? Remember we have **4** engines!

What is the jet power?

What is the jet efficiency?

Exercise 3

Cessna Citation II - A

0.0/2.0 points (ungraded)

Two experienced test pilots from the faculty of Aerospace Engineering at Delft University of Technology are performing measurements in-flight with the Cessna Citation II (see figure). The following data are given:



$$W = 50,000 \text{ [N]}$$

$$S = 30 \text{ [m}^2\text{]}$$

$$C_D = C_{D_0} + k_2 C_L^2$$

$$C_{D_0} = 0.02$$

$$k_2 = \text{unknown}$$

It is **assumed** that the aerodynamics of this aircraft can be represented by a '**one-term**' lift drag polar. The test pilots fly two test points:

Test point 1:

- (Quasi) straight, symmetric and steady flight at 2000 meters altitude.
- Air density $\rho = 1.0065 \text{ [kg/m}^3\text{]}$
- Throttle in idle position ($T = 0 \text{ [N]}$)
- Airspeed $V = 90 \text{ [m/s]}$
- Rate of descent = 6.3 [m/s]

Test point 2:

- (Quasi) straight, symmetric and steady flight at 2000 meters altitude.
- Air density $\rho = 1.0065 \text{ [kg/m}^3\text{]}$
- Throttle set to full power. It is unknown to the pilots how much thrust is generated by the engines at full power.
- Airspeed $V = 90 \text{ [m/s]}$
- Rate of climb = 13.3 [m/s]

Exercise 3

A) For test-point 1; calculate the lift coefficient (C_L) :

B) For test-point 1; calculate the drag coefficient (C_D):

Calculate the value of constant k_2 in the lift-drag polar based on your results in questions A and B. In case you did not obtain the correct answers in question A and question B, use $C_L = 0.3206$ and $C_D = 0.0253$.

Note: this combination is taken from the same lift-drag polar and will result in the same k_2 value as with the correct answers in question A and question B. In essence it could be the result from a test measurement at a different aircraft weight or airspeed.

C) Calculate the value of constant k_2 in the lift-drag polar:

Exercise 3

D) Using the information obtained in test point 2, determine the maximum amount of thrust (T_{max}) in **kN** that can be generated by this aircraft at 2000 meters altitude.

Note: if you did not obtain the correct answers to questions A and B, do not try to answer this question by your incorrect C_L and C_D values or the ones given as alternative in question C because this will then result in an incorrect value for T_{max} . Try to find the solution by calculating the total aerodynamic drag or power required at 90 m/s based on the data provided for test point 1. If you did get correct values, you can check it using this method.

After testing the pilots return to the airport. During approach the aircraft is flying at 1000m ($\rho = 1.1116 \text{ kg/m}^3$), with flaps extended. This means that there is a new lift-drag polar:

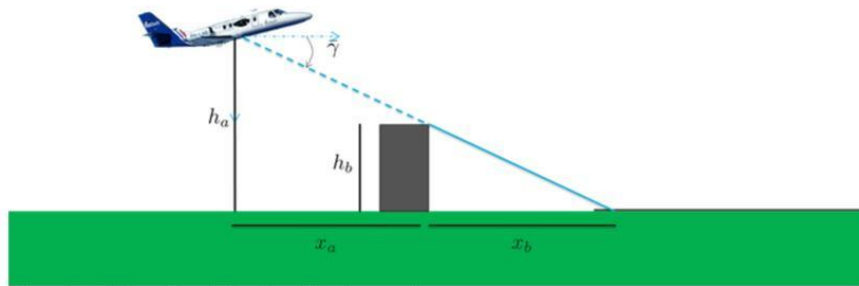
$$C_D = 0.0459 + 0.0370C_L^2$$

The maximum thrust at this altitude is given as $T = 12 \text{ kN}$. Finally, during this approach the aircraft flies at an unknown airspeed, *different* than the airspeed during the test flight.

E) Calculate the lift coefficient (C_L) at which this aircraft should fly to achieve the maximum climb angle (γ_{max}):

Exercise 4

A pilot of an aircraft with engine trouble (so no thrust) wants to land at the nearest airport. Ahead of the airport, there is a building of h_b meters high, as shown in the figure.



The aircraft has the following lift-drag polar:

$$C_{D_0} = 0.02$$

$$k_1 = 0$$

$$k_2 = 0.0352$$

And the geometric data is as follows:

$$x_a = 1000 \text{ m}$$

$$x_b = 715 \text{ m}$$

$$h_a = 119.9 \text{ m}$$

$$h_b = 50 \text{ m}$$

What is the minimum glide angle ($\bar{\gamma}$, so a positive number) of the aircraft? Please indicate your answer in degrees.

 deg